

Capturing traffic over Bluetooth Pairing Sequences and Key Exchange Privacy Analysis of Augmented Reality Glasses

1 Introduction

The captures in this directory was during the initial discovery and pairing mode of the glasses. The pairing was done four times during the trials and was done two times per capture for the channels. The goal for the captures were to hopefully gain some insight about pairing sequences and key exchange.

2 Metadata

Time of each experiment: Nov 4th 21:50
Duration of trial_phone: Roughly 8 minutes
Duration of trial_glasses: Roughly 12 minutes
Duration of channel captures: Roughly 3 minutes

Different addresses:

- FF:FF:FF:FF:FF:FF - as a destination address, is usually for broadcasts of advertising packets or discovery requests. This means that those devices were advertising their presence to nearby devices.
- 98:59:49:34:7A:8C - the static MAC address of the Ray Ban Meta glasses
- B8:3B:CC:DA:73:A8 - the static MAC address of the Redmi phone

3 Problem

There were quite a few broadcast packets from different sources, however, the packets were not enough to determine if an LTK was exchanged. Usually, for an LTK exchange, pairing-specific packets such as Pairing Request, Pairing Response, and/or Encryption Information would be what to look for. In our captures, none of these were evident, which could likely be due to the packets being Malformed.

Due to the same MAC address being displayed everytime on the phone app 'nRF Connect', I believed that the glasses is using a static MAC address for BLE advertising. However, the packets captured called this into question as neither of the MAC addresses of the glasses nor phone appeared in the captures. Problem is that it is suggested to use packet patterns or services (like btcommon services or characteristics associated with the devices) for tracking, however, when I apply the filter in Wireshark, no packets appear.

What I am thinking of doing: Redo experiment again for roughly 10 minutes per captures. Turn Bluetooth on/off bluetooth and try pairing again but this time try and observe in real time if any fluctuations or packets could be the result of those interactions and that the MAC addresses of the device could be randomized MAC addresses used in BLE to prevent tracking.

4 Notes

4.1 trial_for_ltk_phone

This capture had 100% Malformed packets.

None of the source addresses captured was coming from the glasses.

Most packets had FF:FF:FF:FF:FF:FF as a destination address.

4.2 trial_for_ltk_glasses

This capture had 100% Malformed packets.

Most packets had FF as a destination address.

4.3 channel_37

This capture had 99.1% Malformed packets, 1 was not Malformed.

All packets had FF as a destination address.

4.4 channel_38

This capture had 100% Malformed packets.

All packets had FF as a destination address.

4.5 channel_39

This capture had 100% Malformed packets.

Almost all packets had FF as a destination address.